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REVIEW

A Review on Privacy-Preserving Techniques for Spatiotemporal Data

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Abstract

The rapid proliferation of spatiotemporal data—originating from GPS-enabled devices, sensor networks, IoT ecosystems, and social media platforms—has raised critical privacy concerns. Protecting sensitive information while preserving data utility is increasingly becoming a central research challenge. Although existing surveys have addressed privacy-preserving methods, many lack systematic methodological frameworks, rigorous quantitative comparisons, or in-depth analysis of emerging techniques tailored to spatiotemporal contexts. This survey bridges these gaps by offering a comprehensive and structured review of privacy-preserving techniques specific to spatiotemporal data. Our contributions include: (i) a systematic methodology for selecting and evaluating relevant literature from reputable sources, (ii) a novel theoretical classification of techniques including data anonymization, differential privacy, cryptographic methods, federated learning, and geospatial watermarking, (iii) a quantitative comparison based on standardized performance metrics, and (iv) a critical analysis of hybrid approaches that integrate differential privacy, cryptography, and machine learning. Furthermore, the paper highlights current limitations, practical challenges, and key areas for future research, serving as a roadmap for researchers and practitioners. The insights provided aim to foster the development of robust, scalable, and privacy-aware solutions for spatiotemporal data analysis.

Keywords Privacy-preserving techniques · Spatiotemporal data · Differential privacy · Homomorphic encryption · Federated learning

List of Abbreviations

GPS	Global Positioning System
IoT	Internet of Things
IIoT	Industrial Internet of Things
SMC	Secure Multiparty Computation
M-IoT	Mobile Internet of Things
MCS	Mobile Crowdsensing
POIs	Points of Interest
ARS	Area of Responsibility
CA	Classification Accuracy
KL-Divergence	Kullback–Leibler Divergence
IER	Inference Error Rate
SFRR	Statistical Feature Retention Rate
WGAN	Wasserstein Generative Adversarial Network
pCDBN	Probabilistic Conditional Deep Belief Network
STDM	Spatiotemporal Data Mining
HyInc	Hybrid Incentive Mechanism
TPP-DABD	Trajectory Data Publishing by Dynamic Anonymization with Bounded Distortion
SN	Social Network
DPDCM	Differentially Private Data Collection Mechanism
RAPPOR	Randomized Aggregatable Privacy-Preserving Ordinal Response
ORAPPOR	Optimized Randomized Aggregatable Privacy-Preserving Ordinal Response
S2M	Single to Randomized Multiple Dummies
MDN	Multidimensional Negative Surveys

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PDE	Perturbing Data with Sensing Errors
ETE	Estimating True Distribution Considering Sensing Errors
DPCP	Data Privacy and Confidentiality Policy
STMPS	Secure and Trusted Mobile Payment System

1 Introduction

Spatiotemporal data, which integrates both spatial and temporal dimensions, have become a cornerstone for numerous critical applications such as transportation systems, urban planning, healthcare monitoring, and social network analysis [1, 2]. The proliferation of GPS-enabled devices, IoT sensors, and social media platforms continuously generates massive amounts of spatiotemporal data [3]. Harnessing these rich datasets enables stakeholders to derive actionable insights, optimize resource allocation, forecast mobility patterns, and enhance service delivery. However, this growing reliance on spatiotemporal data has also intensified privacy concerns involving unauthorized re-identification, sensitive location inference, and behavioral profiling [4, 5].

Traditional privacy-preserving methods, such as simple anonymization or encryption, are often insufficient for the dynamic and highly granular nature of spatiotemporal data [6, 7]. In addition, many existing survey studies in this domain offer only partial coverage of emerging techniques or lack systematic evaluation frameworks that can account for issues like scalability, computational overhead, and the intricate privacy-utility trade-off [3]. These gaps limit practitioners' ability to select, tailor, or combine the most suitable privacy-preserving strategies for their specific spatiotemporal use cases.

In response, this survey provides a rigorous and structured review of privacy-preserving techniques explicitly designed for spatiotemporal data. We aim to consolidate the growing body of research, critically evaluate existing methods, and identify open challenges that the community must address. Specifically, our investigation is driven by the following research questions:

- **RQ1:** What privacy-preserving techniques are specifically applied to protect spatiotemporal data?
- **RQ2:** What performance metrics and benchmarking approaches are used to evaluate these techniques?
- **RQ3:** What challenges, risks, and research opportunities are identified in the current body of literature?

To answer these questions comprehensively, we carried out a systematic search and selection process (Section 2). We searched leading academic databases for articles published between 2007 and 2023, using carefully chosen keywords such as *"spatiotemporal data privacy"*, *"privacy-preserving techniques"*, and *"location and time-based privacy"*. Each candidate study was then screened against predefined criteria to ensure relevance, methodological rigor, and clear applicability to spatiotemporal contexts. This methodology allowed us to pinpoint both classical and emerging privacy-preserving methods, as well as identify trends in how researchers evaluate these methods' performance.

Contributions and Novelty. The key contributions of this paper include:

- *Systematic Review Methodology:* We adopt established guidelines for literature selection and quality assessment [8], enhancing the transparency and reproducibility of our findings.
- *Comprehensive Classification:* We propose a nuanced taxonomy of privacy-preserving techniques—encompassing data anonymization, differential privacy, cryptographic methods, federated learning, and geospatial watermarking—showcasing how each is tailored or extended for spatiotemporal scenarios [9, 10].
- *In-Depth Comparative Analysis:* We conduct a quantitative comparison across standardized metrics (e.g., privacy leakage, data utility, computational overhead) to elucidate the privacy-utility trade-offs unique to spatiotemporal data [7].
- *Hybrid Approach Insights:* We delve into integrated methods (e.g., differential privacy combined with federated learning) that leverage multiple paradigms for stronger or more scalable privacy guarantees [10].
- *Roadmap for Future Work:* We pinpoint unresolved issues—such as adaptive adversaries, heterogeneous deployment constraints, and the need for unified benchmarking—and provide targeted recommendations to guide ongoing research.

By offering a detailed critique of existing approaches and mapping them to real-world demands, our survey delivers a consolidated resource for practitioners, researchers, and policy-makers interested in preserving the privacy of spatiotemporal data. Furthermore, we highlight key directions for innovation—ranging from advanced cryptographic primitives to context-aware differential privacy schemes—aimed at bridging the gap between strong theoretical guarantees and practical, scalable solutions.

Paper Organization. The remainder of this paper is structured as follows:

- Section 2 outlines our literature selection methodology and filtering criteria.
- Section 3 elaborates on various privacy-preserving techniques and their applications in spatiotemporal contexts.

- Section 4 discusses critical threats, challenges, and adversarial models that compromise spatiotemporal privacy.
- Section 5 presents a comparative evaluation and highlights notable research gaps.
- Section 6 concludes the paper, summarizing key takeaways and sketching a roadmap for future work in privacy-preserving spatiotemporal data analytics.

2 Methodology

To ensure comprehensive coverage and mitigate bias, this review followed a systematic approach to identifying and selecting relevant studies, guided by established guidelines for literature reviews [8]. The filtering criteria and methodology are summarized below:

1. **Search Databases:** We searched major academic databases, including *IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Google Scholar*. These databases were chosen due to their extensive coverage of computer science, data science, and information security publications.
2. **Search Terms and Scope:** Our queries combined domain-specific and privacy-related keywords, such as *"spatiotemporal data," "trajectory data," "location privacy," "privacy-preserving," "differential privacy," "homomorphic encryption,"* and *"federated learning."* Search strings were constructed using Boolean operators (*AND, OR*) to broaden or refine results appropriately. The initial time scope covered *2007-2023*, aligning with the era of major breakthroughs in location-based services and advanced privacy-preserving methodologies.
3. **Inclusion Criteria:**
    - *Relevance:* Studies had to explicitly focus on techniques for privacy preservation in *spatiotemporal* or *location-based* data scenarios.
    - *Peer Review:* Only articles published in *peer-reviewed* journals, conferences, or workshops were included.
    - *Methodological Rigor:* Preference was given to works proposing or empirically evaluating *privacy-preserving algorithms*, frameworks, or architectures.
    - *Data Availability:* Studies providing details on *datasets*, *evaluation metrics*, or *use-case validations* were prioritized.
4. **Exclusion Criteria:**
    - *Irrelevant Focus:* Studies focusing solely on general privacy (e.g., cryptography without spatiotemporal applicability) or purely theoretical papers with no connection to location or trajectory data were excluded.
    - *Duplicate Entries:* Duplicates found in multiple databases were removed.
    - *Short/Non-Archival Papers:* Abstracts, posters, or extended summaries without sufficient methodological details were not considered.
5. **Screening Process:**
    - *Title and Abstract Screening:* Two independent reviewers inspected the titles and abstracts of retrieved articles to remove obviously irrelevant items (e.g., studies outside the domain of spatiotemporal privacy).
    - *Full-Text Review:* Articles that passed the initial screening underwent a *full-text assessment* to verify that they addressed privacy challenges in spatiotemporal data, included rigorous methodology, and provided results or insights applicable to this domain.
    - *Consensus Resolution:* Any disputes regarding an article's inclusion were discussed and resolved via consensus. When necessary, a third reviewer made the final decision.
6. **Final Corpus:** Following the filtering stages, a total of **[4]** articles were selected as the final corpus for this review. These spanned *journal articles*, *conference proceedings*, and *technical reports*, collectively reflecting the breadth of recent developments in privacy-preserving spatiotemporal data management.

This structured selection process ensured a robust and unbiased set of studies, forming the foundation for the subsequent in-depth review of privacy-preserving techniques, comparative evaluations, and identified research gaps.

3 Privacy-Preserving Techniques for Spatiotemporal Data

Privacy-preserving techniques play a pivotal role in protecting sensitive spatiotemporal information while maintaining data utility. As shown in Figure 1, these techniques are categorized into six major classes: Anonymization, Differential Privacy, Cryptographic Techniques, Privacy-Preserving Data Mining and Machine Learning, Location Privacy, and Geospatial Watermarking. Each class comprises specific methods designed to mitigate privacy risks inherent in spatiotemporal data.

These techniques have been applied across various domains including transportation systems, mobility analytics, healthcare monitoring, and smart cities [1–3]. The subsections below provide an in-depth analysis of each category, highlighting their methodologies, use cases, and practical limitations.

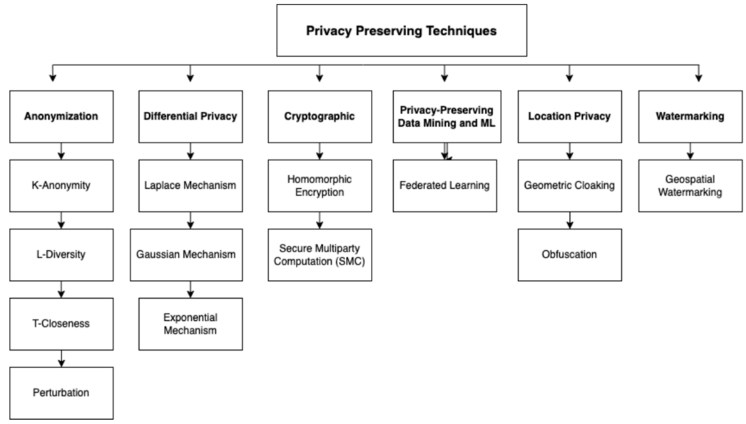

Fig. 1 Taxonomy of Privacy-Preserving Techniques for Spatiotemporal Data

3.1 Anonymization

Anonymization is a foundational technique that transforms datasets in a way that individual identities cannot be distinguished, thus preventing re-identification. Key methods include:

- **k-Anonymity**: Groups records so that each individual is indistinguishable from at least $k - 1$ others. In spatiotemporal datasets, this typically means forming clusters based on location and time features [11].
- **l-Diversity**: Enhances k-Anonymity by ensuring a minimum of l distinct sensitive values within each group [12].
- **t-Closeness**: Extends privacy protection by ensuring the distribution of sensitive attributes in each group is close to the global distribution [13].
- **Generalized Perturbation**: Adds controlled noise or distortion to reduce the risk of re-identification while preserving utility.

These techniques are particularly useful in scenarios involving mobility traces or trajectory publishing. However, they often trade off data granularity for privacy, leading to utility loss [14].

3.2 Differential Privacy (DP)

Differential Privacy (DP) provides formal, mathematical guarantees of privacy by introducing calibrated noise to data queries or outputs. Its adaptability has made it a dominant privacy paradigm for spatiotemporal data [15, 16].

Common DP Mechanisms:

- **Laplace Mechanism**: Adds Laplacian noise based on the sensitivity of a function.
- **Exponential Mechanism**: Useful for non-numeric queries by choosing outputs probabilistically.
- **Gaussian Mechanism**: Adds Gaussian noise, typically applied in high-dimensional data scenarios and (ϵ, δ)-DP frameworks.

DP enables the release of aggregate statistics while preserving user privacy. However, excessive noise can impair spatiotemporal analytics where fine-grained temporal or spatial patterns are needed [7].

3.3 Cryptographic Techniques

Cryptographic approaches enable secure computation without revealing the underlying data. Two major paradigms include:

- **Homomorphic Encryption (HE)**: Allows computations to be performed on encrypted data. Fully Homomorphic Encryption (FHE) supports arbitrary operations but at a high computational cost, while Partially Homomorphic Encryption (PHE) provides a compromise between functionality and efficiency [9].
- **Secure Multiparty Computation (SMC)**: Multiple parties compute a function jointly without revealing individual inputs. This is critical for collaborative spatiotemporal analytics across organizations [10].

These techniques are ideal for use cases like encrypted mobility analysis or secure location-based services.

3.4 Privacy-Preserving Data Mining and Machine Learning

Machine learning models often require large amounts of personal data. Privacy-preserving ML ensures meaningful insights can be extracted without compromising privacy.

- **Federated Learning (FL)**: Enables decentralized training on local devices, transmitting only model updates.

This reduces privacy leakage and has proven effective in spatiotemporal contexts like traffic prediction and location-based health monitoring [10].
- **DP-FL Integration**: Combines federated learning with differential privacy by adding noise to model gradients or updates, ensuring both local data privacy and global model robustness [10].

FL faces practical challenges like communication overhead, model inconsistency, and device heterogeneity, which must be addressed in real-world deployments [17].

3.5 Location Privacy Techniques

Location privacy methods focus on concealing exact user positions and trajectories, a particularly sensitive dimension in spatiotemporal datasets.

- **Geometric Cloaking**: Generalizes location coordinates into broader areas (e.g., bounding rectangles), hiding exact user locations [4].
- **Obfuscation**: Introduces noise or shuffling in location data to prevent tracking while preserving analytical utility [5].

Although effective, such methods may reduce spatial resolution and affect fine-grained analysis, especially in high-precision tasks.

3.6 Geospatial Watermarking

Geospatial watermarking embeds digital signatures or patterns within spatial datasets to verify authenticity and ownership without exposing sensitive content [18].

- **Robust Watermarks**: Resistant to transformations like scaling or compression.
- **Fragile Watermarks**: Detect even small alterations in datasets, ensuring data integrity.

While watermarking is not a privacy-preserving mechanism in the traditional sense, it plays a crucial role in data provenance and secure sharing [18].

4 Threats and Challenges

Despite ongoing advancements, privacy-preserving techniques for spatiotemporal data still face a broad spectrum of technical and practical challenges [12, 18]. These challenges arise from the complex structure of spatiotemporal datasets, real-world deployment scenarios, and the evolving nature of adversarial threats. This section synthesizes critical issues identified across the literature, emphasizing the need to strike a balance between privacy protection and maintaining data utility.

4.1 Vulnerability to Re-identification

Even after applying robust anonymization or obfuscation methods, spatiotemporal datasets remain susceptible to re-identification attacks [11, 12]. This is largely due to the high dimensionality and uniqueness of individual mobility patterns. Linkage attacks, where adversaries use external data sources (e.g., geotagged social media posts or public transit logs), can compromise anonymized datasets. Traditional techniques like k-Anonymity may fail when spatial and temporal quasi-identifiers create uniquely identifiable trajectories within a small geographic or temporal window [13].

4.2 Temporal Privacy Concerns

While many techniques emphasize spatial privacy, fewer directly address temporal dimensions, despite their critical role in user behavior profiling [14, 19]. Repeated observations over time enable adversaries to reconstruct individual routines, identify sensitive visits (e.g., hospitals, religious sites), or infer behavioral trends. Techniques that apply temporal generalization or introduce temporal noise often struggle to maintain analytical precision needed for tasks like real-time monitoring or behavior prediction [18].

4.3 Privacy-Utility Trade-offs

A fundamental challenge in spatiotemporal privacy is balancing protection with data usability. Strong privacy guarantees—such as Differential Privacy with low ϵ values—often lead to significant degradation in data quality [15, 16]. Similarly, overly coarse spatial cloaking or excessive perturbation can reduce interpretability and hinder applications such as traffic optimization, anomaly detection, or epidemiological tracking [7]. Achieving optimal noise calibration or applying adaptive privacy models remains an open research problem.

4.4 Computational Overhead and Real-Time Constraints

Cryptographic methods such as Fully Homomorphic Encryption (FHE) and Secure Multiparty Computation (SMC) offer robust protection, but often incur substantial computational and communication overhead [9, 10]. Their implementation in time-critical or resource-constrained environments like mobile or IoT systems remains challenging. Even federated learning—while decentralized and privacy-conscious—faces practical scalability issues when synchronizing updates across thousands of edge devices [10, 17]. Real-time applications demand lightweight, yet secure alternatives that maintain privacy without compromising responsiveness.

4.5 Heterogeneous Deployment Environments

As spatiotemporal data collection expands across IoT, vehicular networks, and mobile edge systems, deploying uniform privacy mechanisms becomes more difficult [3]. These systems vary in processing power, bandwidth, energy availability, and data formats. Lightweight cryptographic and obfuscation techniques are essential, yet must remain resistant to attacks. Advanced privacy-preserving mechanisms like machine learning-based anonymization or DP often require significant computational resources, posing challenges in embedded or low-power environments [19].

4.6 Inconsistent Evaluation Frameworks

One of the key barriers to standardization is the lack of uniform benchmarking protocols. Studies utilize different datasets (e.g., GeoLife, Gowalla, Actitracker) and inconsistent metrics such as privacy leakage, utility loss, F1-score, Mean Relative Error (MRE), or entropy [10, 13]. Moreover, variations in simulation environments, hardware configurations, and threat models hinder reproducibility and cross-study comparability. A community-agreed evaluation framework is essential for consistent performance assessment and fair comparison of privacy-preserving solutions.

4.7 Evolving Threat Models and Adaptive Adversaries

As privacy techniques evolve, so do the strategies of adversaries. New attacks now incorporate machine learning to exploit residual correlations in anonymized or perturbed data [7]. Inference attacks, trajectory reconstruction, and deep learning-based linkage attacks can breach traditional privacy barriers, particularly when adversaries possess partial background knowledge [18, 19]. Therefore, defensive mechanisms must anticipate adaptive threats and consider worst-case scenarios under varied attacker capabilities.

4.8 Limitations in Specific Techniques

Each privacy-preserving category faces inherent limitations:

- **Anonymization techniques** may result in high information loss, limiting spatiotemporal resolution [12].
- **Differential privacy** introduces statistical noise, which can impair small-sample analyses or rare event detection [16].
- **Cryptographic methods** increase latency and hardware complexity [9].
- **Federated Learning** often suffers from aggregation bottlenecks and privacy leakage via gradient inversion attacks [17].
- **Location privacy methods** like obfuscation and cloaking reduce spatial fidelity and may fail against strong inference models [5].
- **Geospatial watermarking** is vulnerable to removal or tampering attacks if not designed with robustness constraints [18].

4.9 Future Directions

To address these challenges, future research should emphasize:

- Development of adaptive privacy mechanisms that dynamically adjust privacy levels based on context and data sensitivity.
- Integration of multi-layered hybrid approaches (e.g., DP+Crypto+ML) tailored to spatiotemporal data complexity.
- Deployment-friendly frameworks combining low-complexity encryption with scalable aggregation protocols.
- Standardization of evaluation metrics and adoption of open benchmark datasets.
- Modeling of realistic attacker capabilities to future-proof privacy solutions.

Overall, these challenges highlight the ongoing need for scalable, context-aware, and interoperable privacy-preserving solutions designed specifically for the intricacies of spatiotemporal data. Section 5 delves into how various studies have evaluated these issues quantitatively, shedding light on key trade-offs and comparative performance insights across the literature.

5 Discussion: Quantitative Insights, Comparative Evaluation, and Challenges

While each privacy-preserving technique for spatiotemporal data demonstrates unique strengths, they also come with critical limitations and trade-offs that must be addressed for real-world deployment. This section presents an in-depth comparative analysis of these techniques using standardized evaluation metrics, empirical evidence from the literature, and case-based examples. The aim is to consolidate quantitative insights and highlight key gaps and opportunities in the field.

Table 1 Comparative Analysis of Privacy-Preserving Techniques

Technique	Privacy Strength (PS)	Utility Preservation (UP)	Computational Overhead (CO)	Scalability (SC)
Anonymization (k-Anonymity, l-Diversity, t-Closeness)	Medium	Medium	Low–Medium	Medium
Differential Privacy (Laplace, Gaussian, Exponential)	High	Low–Medium (varies with ϵ)	Medium	Medium
Homomorphic Encryption / SMC	Very High	High	High	Low–Medium
Federated Learning (with DP)	Medium–High	High	Medium (Comm. overhead)	High
Location Privacy (cloaking, obfuscation)	Medium	Low–Medium	Low	High
Geospatial Watermarking	Low–Medium	High	Low	High

5.1 Strengths and Limitations of Techniques

Anonymization techniques, including k-Anonymity, l-Diversity, and t-Closeness, provide identity protection by generalizing or suppressing quasi-identifiers. However, these methods face challenges such as vulnerability to background knowledge attacks and potential loss of data utility. The privacy guarantee improves with larger values of k, but this often results in high generalization and reduced utility, especially for high-dimensional spatiotemporal datasets [11–13].

Differential Privacy (DP) offers rigorous privacy guarantees and remains robust against various types of inference attacks. However, the performance of DP is highly sensitive to the privacy budget ϵ. Smaller values of ϵ improve privacy but introduce excessive noise, degrading the analytical quality of the data [7, 15, 16]. Gaussian and Laplace mechanisms remain most commonly used, with Gaussian DP better suited for high-dimensional spatiotemporal settings [18].

Cryptographic techniques, particularly Homomorphic Encryption (HE) and Secure Multiparty Computation (SMC), ensure strong privacy even during computation. However, FHE introduces substantial computational overhead, limiting real-time analytics capabilities. Partially Homomorphic Encryption (PHE) balances utility and performance but restricts supported operations [9, 10].

Federated Learning (FL) provides decentralized privacy protection by training models locally on edge devices. It preserves data utility and avoids centralized breaches but suffers from high communication overhead and the risk of gradient leakage. DP-FL combinations mitigate these risks but add additional computational complexity and affect convergence rates [10, 17].

Location Privacy techniques, such as geometric cloaking and obfuscation, reduce location precision to mitigate disclosure risks. However, these approaches degrade spatial resolution, impacting downstream analysis and affecting navigation, traffic prediction, and service delivery [4, 5].

Geospatial Watermarking techniques enhance data provenance and ownership integrity but offer limited direct privacy guarantees and are vulnerable to removal attacks. Integration with other privacy-preserving techniques is often necessary for robust protection [18].

5.2 Comparative Evaluation: Multi-Criteria Analysis

A structured multidimensional evaluation framework is applied across four axes: Privacy Strength (PS), Utility Preservation (UP), Computational Overhead (CO), and Scalability (SC). Table 1 presents a synthesized comparison based on findings reported in the literature.

5.3 Technique-Specific Challenges and Dataset Usage

To complement our multi-criteria analysis, Table 2 provides a technique-level summary of common flaws and the datasets used in corresponding evaluations. This table highlights how specific challenges are reflected in experimental setups across various studies and underscores dataset selection patterns that may influence the generalizability of findings.

5.4 Mapping Selected Studies to the Proposed Taxonomy

Table 3 illustrates how key studies in our final corpus align with the four major categories that were actually used among them: (**Anonymization**, **DP**, **Privacy-Preserving Machine Learning (PP-ML)**, and **Location Privacy**). We also present

Table 2 Overview of Privacy-Preserving Techniques, Challenges, and Common Evaluation Datasets

Privacy-Preserving Technique	Flaws and Limitations	Common Datasets Used
Anonymization (K-Anonymity, L-Diversity, T-Closeness)	Vulnerable to re-identification; utility loss due to over-generalization; limited scalability in high dimensions	GeoLife GPS Trajectories, Gowalla Dataset
Differential Privacy (Laplace, Gaussian, Exponential)	Accuracy degradation due to noise; balancing privacy-utility trade-offs; hard to tune ϵ	GeoLife, Twitter Geo-Tagged Dataset, Actitracker
Cryptographic Techniques (HE, SMC)	High computational cost; limited real-time application feasibility; complex key management	Typically synthetic or institution-specific datasets; often not publicly shared
Privacy-Preserving Data Mining and Machine Learning	Communication bottlenecks in federated learning; model drift; need for robust aggregation	GeoLife, CRAWDAD, Foursquare, Actitracker
Location Privacy (Geometric Cloaking, Obfuscation)	Data granularity loss; reduced accuracy in services; privacy-utility balance difficult	Foursquare/Swarm, Gowalla, Twitter Location Check-ins
Geospatial Watermarking	Vulnerable to watermark removal; less effective alone; limited contribution to anonymity	Geospatial Imagery Datasets, remote sensing and spatial map datasets

Table 3 Study-to-Taxonomy Mapping of Selected Works

Ref.	Datasets Used	Technique Categories Actually Used				Key Metrics
		Anonym.	DP	PP-ML	LocPriv	
[1]	Industrial applications (unspecified)		✓			Privacy risk, data utility, privacy leakage
[2]	Synthetic data (Laplace noise)		✓			Statistical queries, privacy-accuracy trade-off
[4]	Adult Database, Lands End Database	✓				Generalization height, group size, KL-Divergence
[7]	Real trajectory data (Fuzhou)	✓			✓	Inference Error Rate, Statistical Feature Retention
[13]	Gowalla, Foursquare, Yelp, T-Drive, 311, CRAWDAD	✓			✓	Privacy-preserving data publication
[15]	Not specified		✓			Data utility, performance overhead
[24]	Actitracker dataset			✓		Accuracy, precision, recall, F1 Score
[26]	Trajectories of schools in Pune	✓			✓	Precision, recall (map anonymization)

the primary dataset(s) each work employs and the key evaluation metrics they report.

As Table 3 shows, **Anonymization** and **Differential Privacy** approaches dominate many of the selected studies, often targeting identity protection and formal privacy guarantees, respectively. **Location privacy** (cloaking, obfuscation) remains a common strategy for trajectory-based scenarios. Meanwhile, **privacy-preserving machine learning (PP-ML)** techniques appear in contexts where data utility must be preserved at scale (e.g., sensor data analysis).

Although our taxonomy originally included *Cryptographic Methods* and *Geospatial Watermarking*, none of the final selected studies implemented or evaluated them in a spatiotemporal setting; hence, those columns have been removed here. Nonetheless, both methods remain important in broader privacy literature and may be adopted in future work.

Table 4 Dataset Summary Table

Dataset Name	Number of Samples	Number of Features	Features	Application
GeoLife	24,876,978	5	USER ID, TIMESTAMP, LATITUDE, LONGITUDE, ALTITUDE	Outdoor Activities Tracking in Beijing, China
Gowalla	6,442,890	5	USER ID, TIMESTAMP, LATITUDE, LONGITUDE, LOCATION ID	Social Network Check-ins
Actitracker	1,764	4	PARTICIPANT ID, ACTIVITY ID, ACCELERATION X, ACCELERATION Y, ACCELERATION Z	Human Activity Recognition
STL-10	113,000	5000	IMAGES	Image Recognition
NUS-WIDE	269,648	81	IMAGES	Image Classification
PEMS	48	963	TRAFFIC FLOW DATA	Traffic Flow Prediction
DLEMP	21,209	1,009	WEB USAGE DATA	Web Usage Behavior Analysis
311 Dataset	16,137,635	13	SERVICE REQUEST DATA	Urban Service Request Analysis
Crawdad	Varies	Varies	WIRELESS NETWORK DATA	Wireless Network Research
Lands End Database	4,591,581	8	ZIPCODE, ORDER DATE, GENDER, STYLE, PRICE, QUANTITY, SHIPMENT, COST	Point-of-Sale Information
Adult Database	45,222	9	AGE, GENDER, RACE, MARITAL STATUS, EDUCATION, NATIVE COUNTRY, WORK CLASS, SALARY CLASS, OCCUPATION	Evaluation of L-Diversity
MNIST	60,000	784	IMAGES	Image Classification
YesIWell	1,100	15	WELLNESS DATA	Wellness Data Analysis
COVID-19 Dataset	427,036	23	AGE, SEX, ADMINISTRATIVE DIVISION, DATE OF CONFIRMATION, AND CHRONIC DISEASE STATUS	Used to Analyze Attributes Related to the 2019 Coronavirus Disease (COVID-19)

6 Conclusion and Future Directions

In the era of escalating spatiotemporal data stemming from GPS devices, IoT sensors, and social media platforms, safeguarding individual privacy has emerged as a fundamental challenge. The ever-increasing availability of high-resolution spatiotemporal data has significant implications for advancing sectors such as transportation, urban planning, and healthcare. However, it also intensifies the risks of privacy violations, including location tracking, behavioral profiling, and re-identification attacks.

This survey has provided a systematic, detailed, and quantitative review of privacy-preserving techniques specifically tailored for spatiotemporal data. Building on a rigorous methodological framework, we identified six major categories of techniques—Anonymization, Differential Privacy, Cryptographic Methods, Privacy-Preserving Data Mining and Federated Learning, Location Privacy, and Geospatial Watermarking. Unlike prior surveys that merely enumerated techniques, our review critically evaluated their strengths, limitations, and practical trade-offs using standardized metrics and a multi-criteria comparative analysis.

Key findings from our analysis reveal:

Table 5 Privacy and Utility Evaluation Metrics Table

Evaluation Metric	Definition
Privacy Risk	Measure of the potential risk of privacy breaches or violations associated with a particular data release or processing activity.
Data Utility	Measure of the usefulness or quality of data after applying privacy-preserving techniques.
Privacy Leakage	Measure of the amount or extent of sensitive information that is unintentionally disclosed or leaked through data processing or release.
Statistical Queries	Queries that retrieve aggregated or statistical information from a dataset, often used in the context of differential privacy.
Privacy and Accuracy	Trade-off between preserving privacy and maintaining the accuracy of data or results.
Generalization Height	Level of granularity or detail to which data is generalized or anonymized, often used in k-anonymity and l-diversity techniques.
Average Group Size	Average number of individuals or records within a group or cluster in an anonymized dataset.
Discernibility	Measure of how easily an individual record can be distinguished or re-identified from an anonymized dataset.
KL-Divergence	Measure of how one probability distribution diverges from a second, expected distribution, often used in privacy-preserving data analysis.
Inference Error Rate (IER)	Rate at which an adversary can accurately infer sensitive information from an anonymized dataset, used in trajectory data privacy evaluation.
Statistical Feature Retention Rate (SFRR)	Rate at which important statistical features are retained in an anonymized dataset, used in trajectory data privacy evaluation.
Trust Establishment Time	Time taken to establish trust relationships or levels between entities in a trust management system, used in M-IoT applications.
Communication Overhead	Additional communication resources or bandwidth required to implement a trust management system, used in M-IoT applications.
Scalability	Measure of a system's ability to handle increasing amounts of work or its potential to accommodate growth, often used in evaluating trust management systems.
Resilience to Attacks	Ability of a system to resist or recover from attacks or failures, often used in evaluating the robustness of trust management approaches.
Correlation of Selected Subsets of Observations	Measure of the correlation between subsets of observations selected over time, used in mobile crowdsensing privacy evaluation.
Clustering Strategy Performance	Measure of the effectiveness or efficiency of a clustering strategy in selecting measurements, used in mobile crowdsensing privacy evaluation.
Precision	Measure of the correctness of true positive predictions, often used in binary classification tasks.
Recall	Measure of the completeness of true positive predictions, often used in binary classification tasks.
F1 Score	Harmonic mean of precision and recall, used as a single metric to evaluate the balance between precision and recall in binary classification tasks.

- **Differential Privacy (DP)** offers strong theoretical privacy guarantees, yet its effectiveness hinges on carefully tuning the privacy budget ϵ, which significantly impacts data utility.
- **Cryptographic Techniques**, such as Homomorphic Encryption and Secure Multiparty Computation, ensure high levels of confidentiality but remain computationally intensive and may not scale efficiently in real-time or large-scale deployments.
- **Federated Learning (FL)** represents a promising paradigm for decentralized learning without raw data sharing. However, challenges such as communication overhead, model convergence delays, and vulnerability to inference attacks require further optimization.
- **Location Privacy and Watermarking** methods are effective against specific threats—such as fine-grained location disclosure and data tampering—but they offer limited protection against broader privacy risks like re-identification and attribute inference.

Table 6 Summary of Privacy-preserving Techniques and Evaluations

	Privacy Technique	Dataset Details	Evaluation Metrics	Contributions/Key Findings
[1]	Differential Privacy	Industrial applications datasets	Privacy risk, data utility, privacy leakage	Evaluation of DP mechanisms in industrial applications
[2]	Differential Privacy	Synthetic datasets, Laplace noise	Statistical queries, privacy and accuracy	Evaluation of DP algorithms using Laplace noise
[3]	Not specified	Smart grid multidimensional data	Security, efficiency	Proposed privacy-preserving data aggregation scheme
[4]	K-Anonymity, l-Diversity	Adult Database, Lands End Database	Generalization height, average group size, discernibility, KL-divergence	Evaluation of l-diversity method using Incognito
[7]	Anonymization	Real-world trajectory datasets	Inference Error Rate (IER), Statistical Feature Retention Rate (SFRR)	Evaluation of trajectory data privacy protection approach
[8]	Trust Management	Not specified	Trust establishment time, communication overhead, scalability, resilience to attacks	Evaluation of trust management approaches for M-IoT
[6]	Various privacy-preserving techniques	Not specified	Not specified	Provided a comprehensive survey of privacy techniques
[7]	Spatial–temporal cloaking	Real-world trajectory dataset obtained from a volunteer's one-day journey in Fuzhou	Not specified	Proposed secure and efficient privacy-preserving mechanism
[8]	Diverse privacy mechanisms	Not specified	Not specified	Proposed diverse privacy mechanisms
[9]	Distributed privacy-preserving aggregation	Not specified	Not specified	Proposed distributed privacy-preserving aggregation
[10]	Diverse privacy-preserving methods	Not specified	Not specified	Proposed diverse privacy-preserving methods
[11]	Privacy-preserving data analysis	Not specified	Not specified	Proposed mechanisms for secure privacy-preserving analysis
[12]	Mobile crowd sensing	Not specified	Not specified	Proposed privacy-preserving data collection mechanism
[13]	Anonymization and Obfuscation techniques	Synthetic datasets and real-world open datasets, including check-in data from Gowalla, Foursquare, Yelp, trajectory data from T-drive, Beijing taxis trajectory, 311 dataset, CRAWDAD, and SensorScope Data	Not specified	Proposed mechanisms for privacy-preserving publication
[14]	Anonymization	Not specified	Privacy Preservation, Data Utility, Performance Overhead	Proposed l-diversity technique
[15]	Differential Privacy	Not specified	Data Utility, Performance Overhead	Proposed t-closeness technique

Table 6 continued

	Privacy Technique	Dataset Details	Evaluation Metrics	Contributions/Key Findings
[16]	Privacy-preserving data mining techniques	Not specified	Anonymity Level, Data Utility, Spatial–Temporal Accuracy, Performance Overhead	Conducted survey of differential privacy techniques
[18]	Spatiotemporal data publishing	GeoLife, Gowalla datasets	Haversine distance, compression ratios	Evaluation of privacy-preserving framework for spatiotemporal data
[19]	Spatiotemporal data privacy	GeoLife, Gowalla datasets	Utility, file size, spatial compression ratios	Evaluation of privacy-preserving framework for spatiotemporal data
[20]	Mobile Crowdsensing	Synthetic datasets, real-world datasets (e.g., 311 dataset, CRAWDAD)	Correlation of selected subsets of observations, clustering strategy performance	Evaluation of privacy-preserving framework for spatiotemporal data
[21]	ML and DL methods	Various datasets (STL-10, NUS-WIDE, PeMS, DLeMP, etc.)	Prediction accuracy, resource consumption	Evaluates ML and DL methods for network security, focusing on prediction accuracy and resource consumption using various datasets
[22]	Differential Privacy	GeoLife version 1.3, Gowalla	Haversine distance, file size compression ratio, spatial compression ratio	Comparison of baseline Laplace mechanism with temporal hierarchy framework for differential privacy in spatiotemporal data
[23]	Differential Privacy	Not provided	MRE, Pearson Correlation	Evaluation of a spatiotemporal density release framework with differential privacy, demonstrating practical utility with strong privacy guarantees
[24]	Federated Learning	Actitracker dataset	Accuracy, Precision, Recall, F1 Score	Evaluation of CROWDFL, a privacy-preserving mobile crowdsensing system, using the Actitracker dataset for human activity recognition, demonstrating effectiveness and efficiency through theoretical analysis and experimental testing
[26]	Anonymization	Trajectories of schools in Pune	Precision, Recall	Experimental evaluation of a privacy-preserving technique for static data release using map anonymization, demonstrating effectiveness through precision and recall metrics on real map data

Future Research Directions:

1. **Standardized Benchmarks and Evaluation Protocols:** There is a pressing need for common publicly available spatiotemporal datasets, unified evaluation metrics, and reproducible benchmarking protocols to facilitate fair comparison and accelerate innovation.
2. **Context-Aware Differential Privacy Mechanisms:** Future work should explore adaptive DP models that dynamically adjust noise addition based on data sensitivity, query context, or temporal–spatial granularity, thereby minimizing unnecessary utility loss.
3. **Scalable Cryptographic Frameworks:** Developing lightweight or hybrid encryption models—such as partially homomorphic or approximate encryption—could enable secure analytics with reduced latency and computational burden, especially for IoT and edge computing environments.
4. **Integrated Privacy-Preserving Architectures:** Combining FL, DP, cryptographic methods, and anonymization in a cohesive and interoperable framework could mitigate limitations of individual approaches. This requires the development of efficient coordination protocols and rigorous security proofs.
5. **Explainable Privacy and Usability Tools:** Future systems must provide interpretability and user control-enabling stakeholders to understand the implications of privacy settings (e.g., ϵ values, anonymization thresholds) on data accuracy and utility. User-centric dashboards and explainable privacy interfaces could play a vital role in bridging the technical gap.

As spatiotemporal data continue to proliferate, future research must prioritize adaptable, scalable, and user-aligned privacy solutions. The insights and comparative evaluations presented in this survey offer a structured roadmap for researchers and practitioners striving to develop more secure, efficient, and ethically responsible approaches to spatiotemporal data privacy.

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Declarations

Ethical Approval and Consent to Participate The study was conducted in accordance with ethical standards, and consent was obtained from all participants.

Consent for Publication Consent for publication has been obtained from all co-authors and from the responsible authorities at the institute/organization where the work has been carried out.

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